

* Module 5 → Magnetic Circuits:

(i) Introduction Magnetic Ckts:

When electric current flows through a conductor, a magnetic field is produced around it.

• This is the basic magnetic effect of electric current

• If conductor is wound into a coil, the magnetic field becomes stronger.

• If the above coil is wound on a magnetic material like iron, the magnetic field gets concentrated and follows a definite closed path.

This closed path followed by magnetic flux is called magnetic circuits.

Examples:

- Transformer core • Motor stator • generator core
- Electromagnet • Relay coil.

(ii) Basic Magnetic quantities:

1. Magnetic flux (Φ):

• It is the total magnetic field passing through a surface.

• It Represents the total number of magnetic lines of force linked with the magnetic path.

• Φ = Magnetic flux

• SI Unit = Weber (Wb)

• Flux tells how much total magnetic field is present in the magnetic circuit.

2. Flux Density (B):

Flux density is the magnetic flux per unit area normal to the field.

$$B = \frac{\Phi}{A} ; \text{SI} \rightarrow \text{Tesla (T)} = \text{wb/m}^2$$

• A = Cross sectional Area of the core

- **Meaning:** Flux density tells us how concentrated the magnetic flux is.

If the same flux passes through a smaller area, the flux density becomes larger.

3. Magnetic field intensity (H):

Magnetic field intensity is the magnetizing force that produces magnetic flux in a magnetic circuit.

$$H = \frac{NI}{l} \quad \text{SI unit, A/m}$$

Where,

N = number of turns

I = current through the coil

l = mean length of magnetic path.

H $\Rightarrow$ tells how strongly the coil is trying to magnetize the core.

4. Permeability (μ):

Permeability is the property of a material that indicates how easily it allows magnetic flux to be established in it.

$$\mu = \mu_0 \times \mu_r$$

Where,

$\mu_0 \rightarrow$ permeability of free space.

$\mu_r \rightarrow$ relative permeability of the material.

Meaning $\rightarrow$ It tells us how well a material responds to the magnetizing force (H).

- For same value of (H)

- Air gives small magnetic effect

- Iron gives large magnetic effect.

So, Iron has high permeability, while air has very low permeability.

5. Magnetomotive force (MMF):

Magnetomotive force is the magnetic driving force that establishes magnetic flux in a magnetic ckt.

$$MMF = NI$$

SI unit $\rightarrow$ Ampere-turns (AT)

Meaning $\rightarrow$ MMF plays the same role in magnetic ckt as EMF does in electric ckt. MMF is the cause, and the magnetic flux is the effect.

6. Reluctance (R):

Reluctance is the opposition offered by a magnetic ckt. to the establishment of magnetic flux.

$$R = \frac{l}{\mu A}$$

; SI unit
AT/Wb

- Reluctance is the magnetic equivalent of electrical resistance.
- Magnetic material with high permeability offer low reluctance.

• * Relation b/w H , B and ϕ :

- Coil with N turns carrying current I

$$\downarrow$$
$$MMF = NI$$

$$\downarrow$$
$$H = NI/l$$

(magnetizing force per unit length)

$$\downarrow$$
$$B = \mu H = \mu_0 \mu_r H$$

(flux density produced in the material)

$$\downarrow$$
$$\phi = BA$$

(total magnetic flux in the core)

- So, ϕ from above,
- $$\phi = \mu H A$$

$$\rightarrow \text{subs. } H = \frac{NI}{l}$$

$$\rightarrow \phi = \mu \left(\frac{NI}{l} \right) A$$

$$\phi = \frac{\mu ANI}{l}$$

So, from here from above ϕ

$$R = \frac{l}{\mu A}$$

$$\text{So, } \phi = \frac{\text{MMF}}{R} = \frac{NI \rightarrow \text{MMF}}{\left(\frac{l}{\mu A} \right) \rightarrow R}$$

$$\Rightarrow \boxed{\phi = \frac{\text{MMF}}{R}}$$

$$\begin{aligned} \mu &= \mu_0 \times \mu_r \\ \text{and } &= \mu_0 \times \mu_r \\ \mu_0 &= 4\pi \times 10^{-7} \end{aligned}$$

* Where,

• $\mu \rightarrow$ absolute permeability of material
(H/m)

• $\mu_0 \rightarrow$ permeability of free space (H/m)

• $\mu_r \rightarrow$ relative permeability
(No unit) As,

$$\mu_r = \frac{\mu}{\mu_0} = \frac{(\text{H/m})}{(\text{H/m})} \rightarrow \text{dimensionless}$$

* Energy stored in the magnetic field.

$$\phi = Li; \quad e = \frac{d\phi}{dt} = -L \frac{di}{dt}.$$

Applied voltage must balance the voltage drop across resistor, 'R', and neutralise the above induced emf.

$$V = iR + L \frac{di}{dt}$$

Multiplying throughout by $i \cdot dt$;

$$V i dt = i^2 R dt + L i di$$

↳ Energy supplied by the source in time dt .

↳ $i^2 R dt$, energy dissipated in the form of heat

↳ $L i di \rightarrow$ Energy absorbed by the inductance of the coil in building up the magnetic field. in (joules)

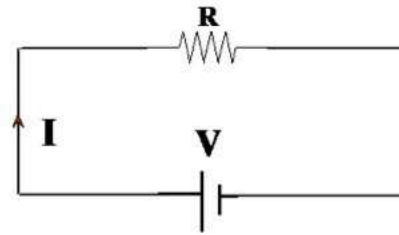
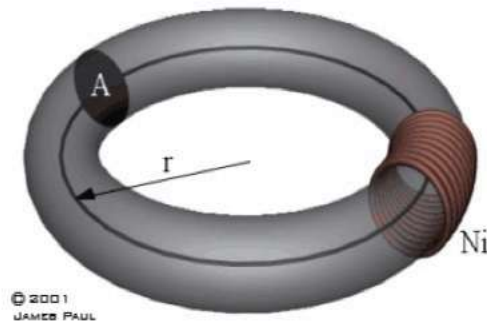
So, Total energy absorbed by the magnetic field when current increases from zero to 1 Ampere

$$= \int_0^1 L i di = L \int_0^1 i di = \boxed{\frac{1}{2} L i^2}$$



Magnetic circuit

The complete closed path followed by any group of magnetic lines of flux



Equivalent electrical circuit

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Magnetic circuit

Analogy with Electric circuits

Similarities:			
Electric circuit		Magnetic circuit	
Quantity	Unit	Quantity	Unit
EMF ($E=IR$)	Volt (V)	MMF ($F=\phi S$)	Ampere-turns
Current (I)	Ampere (A)	Flux (ϕ)	Weber (Wb)
Current density (J)	A/ m ²	Flux density (B)	Wb / m ² or Tesla
Resistance (R)	Ohm (Ω)	Reluctance (S)	Ampere-turns/Wb
Electric field strength (E)	Volts/m	Magnetic field strength (H)	Ampere-turns/m
Conductivity (σ) $\sigma=l/RA$	Siemen/m	Permeability, μ $\mu=l/SA$	Henry/m

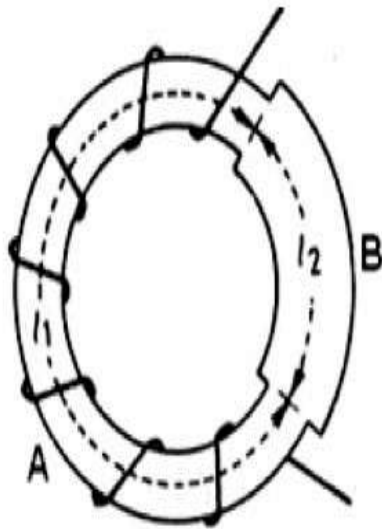
'l' is the length and 'A' is the area of cross section of the conductor

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Composite Magnetic Circuit

Case 1 :



$$\mathcal{R}_1 = \frac{l_1}{\mu_1 A_1}$$

$$\mathcal{R}_2 = \frac{l_2}{\mu_2 A_2}$$

$$\therefore \text{Total Reluctance, } \mathcal{R} = \mathcal{R}_1 + \mathcal{R}_2 = \frac{l_1}{\mu_1 A_1} + \frac{l_2}{\mu_2 A_2}$$

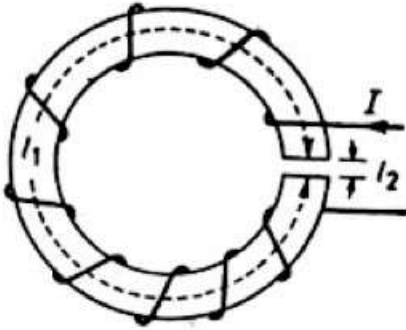
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$$\begin{aligned} \therefore \text{Total flux, } \Phi &= \frac{\text{mmf of coil}}{\text{total reluctance}} \\ &= \frac{F}{\mathcal{R}} = \frac{NI}{\frac{l_1}{\mu_1 A_1} + \frac{l_2}{\mu_2 A_2}} \end{aligned}$$

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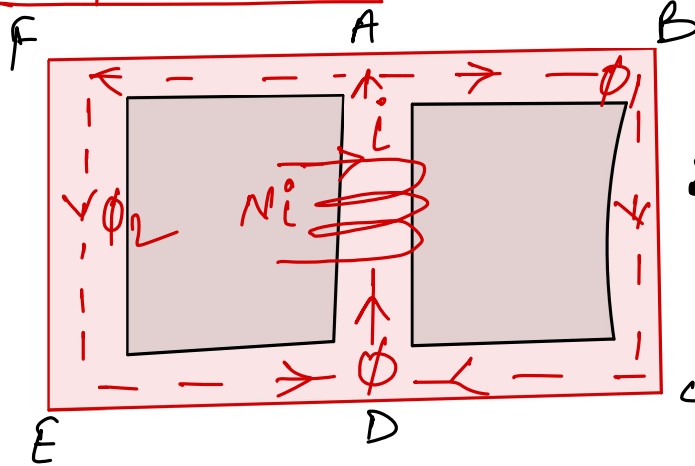
Case 2: (with air gap)



Total reluctance,

$$\begin{aligned}\mathcal{R} &= \frac{l_1}{\mu_1 A} + \frac{l_2}{\mu_0 A} \\ &= \frac{1}{\mu_0 A} \left(\frac{l_1}{(\mu_1 / \mu_0)} + l_2 \right) \\ &= \frac{1}{\mu_0 A} \left(\frac{l_1}{\mu_r} + l_2 \right)\end{aligned}$$

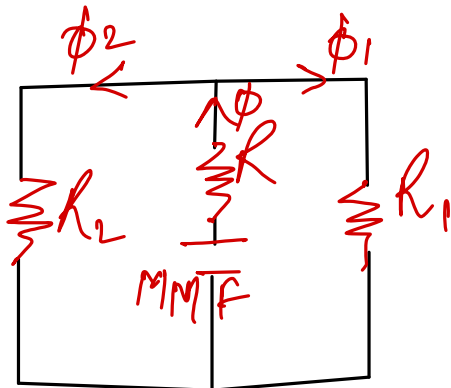
CASE $\rightarrow 3$



So, $\phi = \phi_1 + \phi_2$

Total MMF =
MMF for path AD
+ MMF for path
C ABCD or AFED

So, $\mathcal{R} = \frac{l_c}{\mu_0 \mu_r c A_c}$; $\mathcal{R}_1 = \frac{l_1}{\mu_0 \mu_{r1} A_1}$; $\mathcal{R}_2 = \frac{l_2}{\mu_0 \mu_{r2} A_2}$



$$\begin{aligned}\mathcal{R}_{total} &= \mathcal{R} + \mathcal{R}_1 \parallel \mathcal{R}_2 \\ &= \mathcal{R} + \frac{\mathcal{R}_1 \mathcal{R}_2}{\mathcal{R}_1 + \mathcal{R}_2}\end{aligned}$$

$$\phi = \frac{\text{MMF}}{\text{Reluctance}} = \frac{N I}{\mathcal{R}_{total}}$$

* Fringing or Bulging in a magnetic ckt:

Fringing means when magnetic flux crosses an air gap, the flux lines don't stay perfectly straight and confined to the exact core cross section. They spread outward near the edges of the air gap. This spreading is also called bulging.

So, in the air gap, the effective area carrying flux becomes slightly larger than the actual cross sectional area of the iron core.

→ Iron has very high permeability while air has very low. When flux leaves iron and enters air, it meets much higher reluctance, so the flux lines spread out to find an easier path.

Example

Suppose air gap length $l_g = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

• Core Area
 $A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$

→ without fringing:

$$R_g = \frac{l_g}{\mu_0 A} = \frac{2 \times 10^{-3}}{4\pi \times 10^{-7} \times 4 \times 10^{-4}}$$

$$R_g \approx 3.98 \times 10^6 \text{ AT/Wb}$$

→ with fringing

If effective area becomes say,

$$A_{\text{eff}} = 4.5 \times 10^{-4} \text{ m}^2$$

$$\text{then } R_g = \frac{2 \times 10^{-3}}{4\pi \times 10^{-7} \times 4.5 \times 10^{-4}}$$

$$R_g \approx 3.54 \times 10^6 \text{ AT/Wb}$$

So, Reluctance decreases when fringing is considered, and flux increases.



Q. An iron ring with a cross-sectional area of 3 cm^2 and a mean circumference of 15 cm is wound with 250 turns of wire carrying a current of 0.3 A. The relative permeability of the ring is 1500.

1. Calculate the flux established in the ring.
2. If a saw cut of width 2 mm is made in the above ring, find the new value of the flux in the magnetic circuit.

Solution:

Given,

$$A = 3 \text{ cm}^2 = 3 \times 10^{-4} \text{ m}^2$$

$$l = 15 \text{ cm} = 0.15 \text{ m}$$

$$N = 250, \quad I = 0.3 \text{ A}, \quad \mu_r = 1500$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

(i) Flux established in the ring

MMF is

$$NI = 250 \times 0.3 = 75 \text{ AT}$$

Reluctance of iron ring is

$$R = \frac{l}{\mu_0 \mu_r A}$$
$$R = \frac{0.15}{(4\pi \times 10^{-7})(1500)(3 \times 10^{-4})}$$

$$R = 2.6526 \times 10^5 \text{ AT/Wb}$$

Therefore, flux is

$$\phi = \frac{NI}{R}$$

$$\phi = \frac{75}{2.6526 \times 10^5}$$

$$\phi = 2.83 \times 10^{-4} \text{ Wb}$$

$$\therefore \phi = 2.83 \times 10^{-4} \text{ Wb}$$

(ii) Flux when a saw cut of width 2 mm is made

Air gap length is

$$l_g = 2 \text{ mm} = 0.002 \text{ m}$$

Remaining iron path length is

$$l_{iron} = 0.15 - 0.002 = 0.148 \text{ m}$$

Reluctance of iron portion is

$$R_{iron} = \frac{l_{iron}}{\mu_0 \mu_r A}$$

$$R_{iron} = \frac{0.148}{(4\pi \times 10^{-7})(1500)(3 \times 10^{-4})}$$

$$R_{iron} = 2.6172 \times 10^5 \text{ AT/Wb}$$

Reluctance of air gap is

$$R_g = \frac{l_g}{\mu_0 A}$$

$$R_g = \frac{0.002}{(4\pi \times 10^{-7})(3 \times 10^{-4})}$$

$$R_g = 5.305 \times 10^6 \text{ AT/Wb}$$

Total reluctance is

$$R_T = R_{iron} + R_g$$

$$R_T = 2.6172 \times 10^5 + 5.305 \times 10^6$$

$$R_T = 5.56672 \times 10^6 \text{ AT/Wb}$$

Hence, new flux is

$$\phi' = \frac{NI}{R_T}$$

$$\phi' = \frac{75}{5.56672 \times 10^6}$$

$$\phi' = 1.35 \times 10^{-5} \text{ Wb}$$

$$\therefore \phi' = 1.35 \times 10^{-5} \text{ Wb}$$

Mutual Inductance

When two inductors (or coils) are in a close proximity to each other, the magnetic flux caused by current in one coil links with the other coil, thereby inducing voltage in the latter. This phenomenon is known as *mutual inductance*.

A coil with N turns, when current i flows through the coil, a magnetic flux is produced around it. According to Faraday's law:

$$v = N \frac{d\phi}{dt}$$

Any change in ϕ is caused by a change in the current.

$$v = N \frac{d\phi}{di} \frac{di}{dt}$$

or

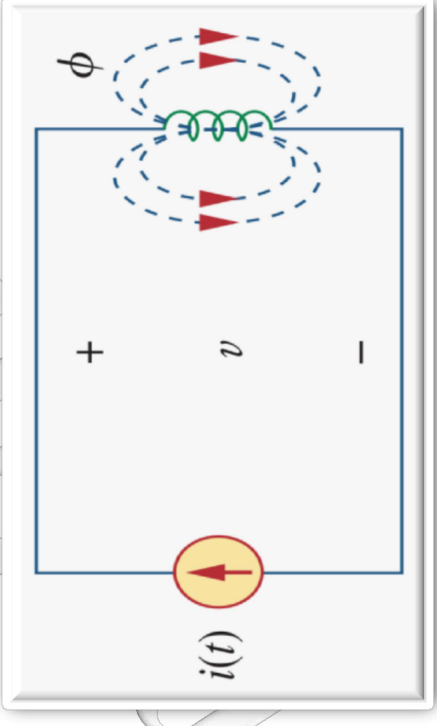
$$v = L \frac{di}{dt}$$

where

$$L = N \frac{d\phi}{di}$$

L is called *self-inductance*, because it relates the voltage induced in a coil by a time-varying current in the same coil.

When two coils with self-inductances L_1 and L_2 that are close to each other. Coil 1 has N_1 turns, while coil 2 has N_2 turns. Assume that the second inductor carries no current. The magnetic flux ϕ_1 emanating from coil 1 has two components: One component ϕ_{11} links only coil 1, and another component ϕ_{12} links both coils.



$$\phi_1 = \phi_{11} + \phi_{12}$$

Mutual Inductance

Entire flux ϕ_1 links coil 1, so the voltage induced in coil 1 is

$$v_1 = N_1 \frac{d\phi_1}{dt} \quad \text{or} \quad v_1 = N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} = L_1 \frac{di_1}{dt}$$

Where $L_1 = N_1 d\phi_1 / di_1$ is the *self inductance* of coil 1

Only flux ϕ_{12} links coil 2, so the voltage induced in coil 2 is

$$v_2 = N_2 \frac{d\phi_{12}}{dt} \quad \text{or} \quad v_2 = N_2 \frac{d\phi_{12}}{di_1} \frac{di_1}{dt} = M_{21} \frac{di_1}{dt}$$

Where

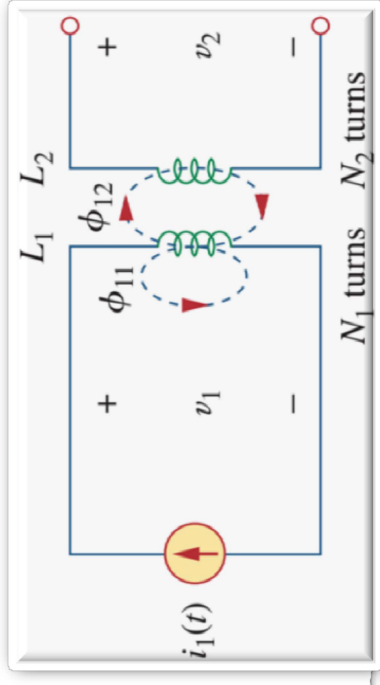
$$M_{21} = N_2 \frac{d\phi_{12}}{di_1}$$

M_{21} is known as the *mutual inductance* of coil 2 with respect to coil 1. Subscript 21 indicates that the inductance relates the voltage induced in coil 2 to the current in coil 1.

The open-circuit mutual voltage (or induced voltage) across coil 2 is

$$v_2 = M_{21} \frac{di_1}{dt}$$

Suppose the current flow in coil 2, while coil 1 carries no current. The magnetic flux ϕ_2 emanating from coil 2 comprises flux ϕ_{22} that links only coil 2 and flux ϕ_{21} that links both coils.



Mutual Inductance

$$\phi_2 = \phi_{21} + \phi_{22}$$

The entire flux ϕ_2 links coil 2, so the voltage induced in coil 2 is

$$v_2 = N_2 \frac{d\phi_2}{dt} = N_2 \frac{d\phi_{21}}{dt} + N_2 \frac{d\phi_{22}}{dt} = L_2 \frac{di_2}{dt}$$

Where $L_2 = N_2 d\phi_2 / di_2$ is the *self inductance* of coil 2

Only flux ϕ_{21} links coil 1, so the voltage induced in coil 1 is

$$v_1 = N_1 \frac{d\phi_{21}}{dt} = N_1 \frac{d\phi_{21}}{di_2} \frac{di_2}{dt} = M_{12} \frac{di_2}{dt}$$

Where

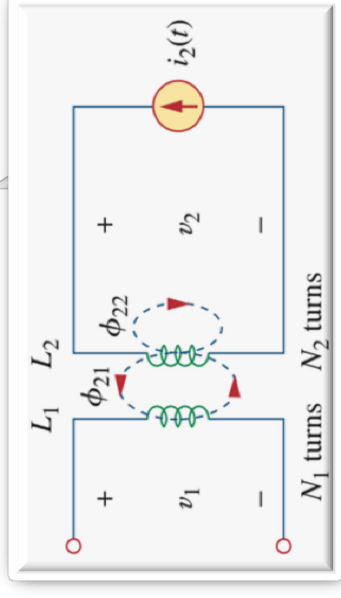
$$M_{12} = N_1 \frac{d\phi_{21}}{di_2}$$

M_{12} is known as the mutual inductance of coil 1 with respect to coil 2.

The open-circuit mutual voltage (or induced voltage) across coil 1 is

$$v_1 = M_{12} \frac{di_2}{dt}$$

$$M_{12} = M_{21} = M$$

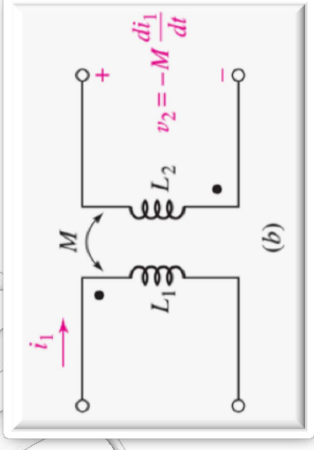
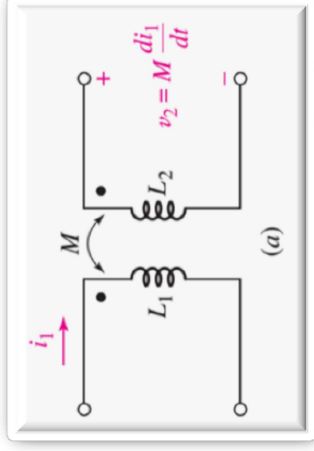


Mutual inductance is the ability of one inductor to induce a voltage across a neighboring inductor, measured in henrys (H).

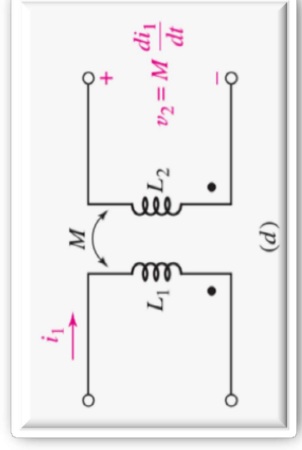
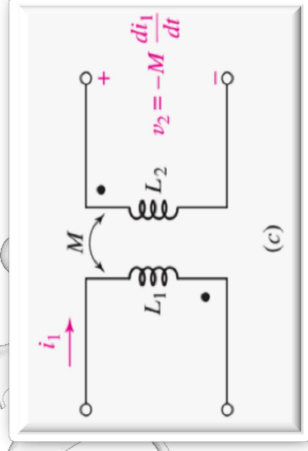
Mutual Inductance - Dot Convention

Although mutual inductance M is always a positive quantity, the mutual voltage $M \frac{di}{dt}$ may be negative or positive. The dots are used along with the dot convention to determine the polarity of the mutual voltage.

If a current enters the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is **positive** at the dotted terminal of the second coil.



If a current leaves the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is **negative** at the dotted terminal of the second coil.



Analyze Circuits Involving Mutual Inductance

The total inductance for coupled coils in series is

$$L = L_1 + L_2 + 2M \quad (\text{Series-aiding connection})$$

$$L = L_1 + L_2 - 2M \quad (\text{Series-opposing connection})$$

For the time domain circuit shown in fig.(a). Applying KVL to coil 1 gives

$$v_1 = i_1 R_1 + L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

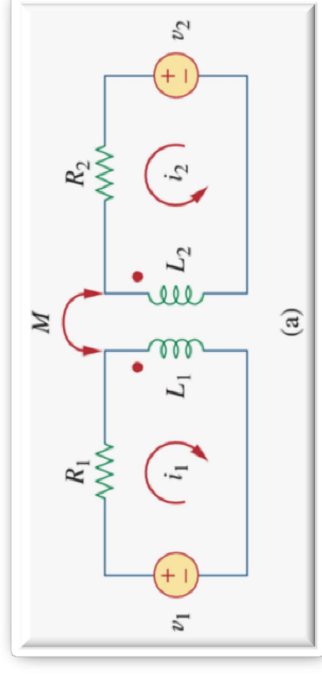
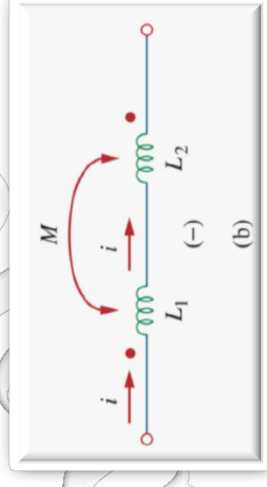
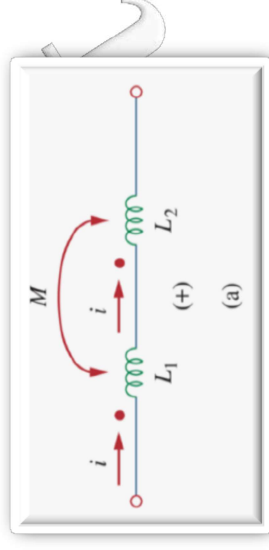
For coil 2, KVL gives

$$v_2 = i_2 R_2 + L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

The above two equations can be written in the frequency domain as

$$V_1 = (R_1 + j\omega L_1)I_1 + j\omega M I_2$$

$$V_2 = j\omega M I_1 + (R_2 + j\omega L_2)I_2$$



Energy in a Coupled Circuit

The Coupling Coefficient

Coupling coefficient (k) is used to describe the degree of coupling between coils.

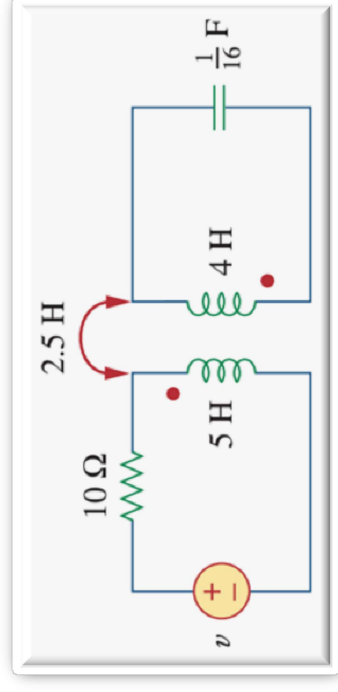
$$M = k\sqrt{L_1 L_2}$$

where $0 \leq k \leq 1$ or equivalently $0 \leq M \leq \sqrt{L_1 L_2}$

- If the entire flux produced by one coil links another coil, then $k=1$ and we have 100 percent coupling, or the coils are said to be *perfectly coupled*.
- For $k < 0.5$, the coils are said to be *loosely coupled*.
- For $k > 0.5$, the coils are said to be *tightly coupled*.

Example: Consider the circuit. Determine the coupling coefficient. Calculate the energy stored in the coupled inductors at time $t=1$ s if $v=60 \cos(4t + 30^\circ)$ V.

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{2.5}{\sqrt{20}} = 0.56$$



Eddy Current Loss

(a) Eddy Current Loss

When a magnetic material (like the core of a transformer or motor) is subjected to a **time-varying magnetic flux**, circulating currents are induced within the material itself. These are called **eddy currents**.

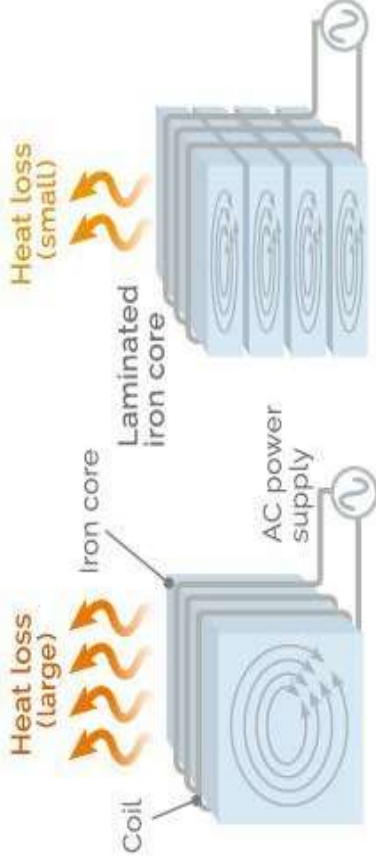
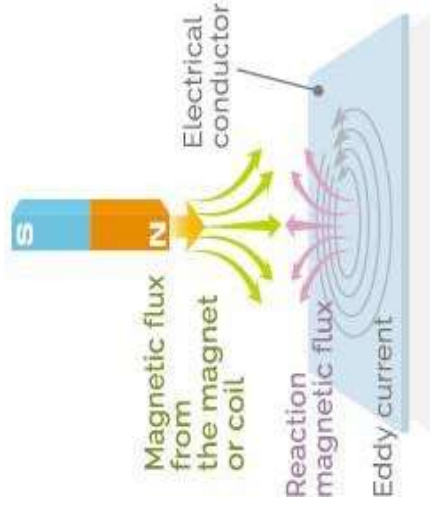
Why they occur

According to **Faraday's Law of Electromagnetic Induction**, a changing magnetic flux induces an EMF. In a bulk conductor, this EMF drives closed-loop currents inside the material.

Effects

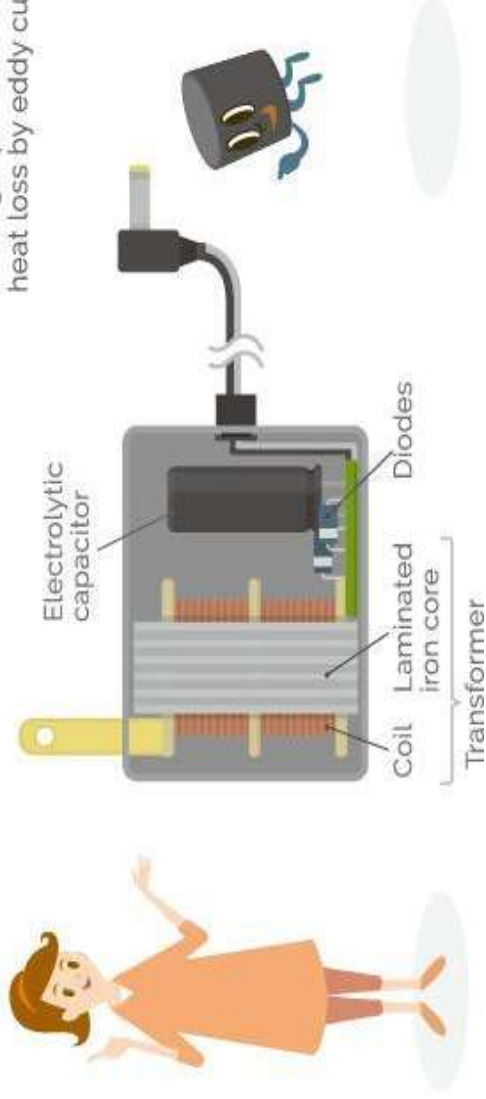
- These circulating currents flow within the core material.
- Since the material has electrical resistance, they produce **I^2R losses**.
- Energy is dissipated as **heat**, reducing efficiency.

< Heat loss due to the iron core of a transformer and eddy current >



Since an iron core has low electrical resistance, it leads to large Joule heat loss by eddy current.

The use of a laminated iron core lengthens the entire route of eddy current and increases the resistance making it possible to reduce heat loss by eddy current.



Mathematical Expression

Eddy current loss is approximately given by:

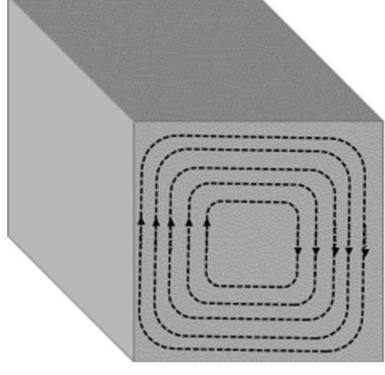
$$P_e \propto B_{max}^2 f^2 t^2$$

Where:

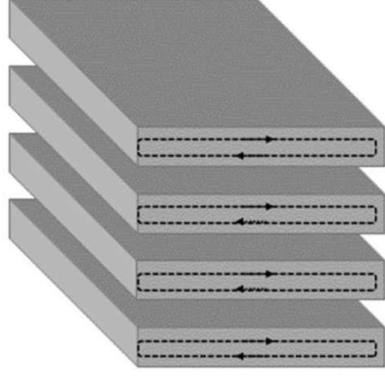
- B_{max} = maximum flux density
- f = frequency
- t = thickness of the core

Methods to Reduce Eddy Current Loss

- Use **laminated cores** (thin insulated sheets)
- Use materials with **high resistivity** (e.g., silicon steel)
- Reduce thickness of laminations



(a) Eddy current loop



(b) Eddy current loop through lamination

Hysteresis Loss

Hysteresis Loss

Hysteresis loss occurs due to the repeated **magnetization and demagnetization** of a magnetic material when subjected to an alternating magnetic field.

Physical Explanation

Magnetic materials are made of **domains**. When an external magnetic field is applied:

- Domains align with the field.
- When the field reverses, domains must realign again.

This realignment is not perfectly reversible, leading to energy loss.

Hysteresis Loop

The relationship between magnetic field intensity (H) and flux density (B) forms a loop called the **Hysteresis Loop**.

- The **area of the loop** represents energy loss per cycle.

Mathematical Expression

$$P_h = \eta B_{max}^n f$$

Where:

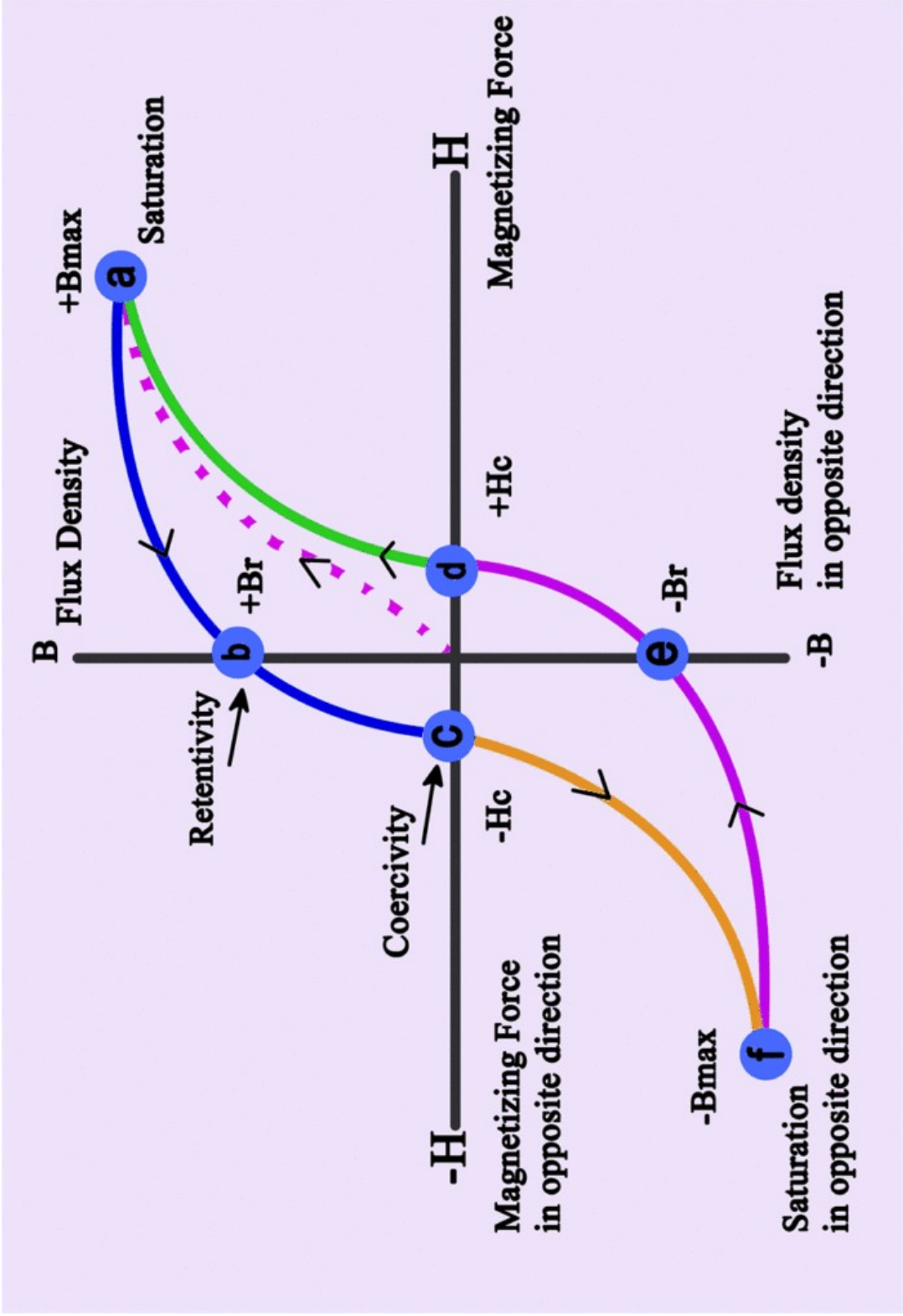
- η = Steinmetz coefficient
- B_{max} =maximum flux density
- f = frequency
- $n \approx 1.6$ (for most materials)

Characteristics

- Depends on material properties
- Increases with frequency
- Causes heating of the core

Reduction Methods

- Use materials with **narrow hysteresis loops** (e.g., silicon steel, ferrites)



The figure shown is the **B–H hysteresis loop** of a ferromagnetic material, which represents the relationship between the **magnetizing force H** applied to a magnetic material and the resulting **magnetic flux density B** produced in it during a complete cycle of magnetization. This loop is formed because the magnetic flux density does not follow the magnetizing force instantaneously; instead, it **lags behind** due to the internal molecular friction and domain realignment effects inside the magnetic material. This lagging phenomenon is known as **magnetic hysteresis**, and the closed curve obtained is called the hysteresis loop.

The horizontal axis of the graph represents the **magnetizing force H** , which is the magnetic field intensity applied to the material and is produced by current flowing through the exciting coil wound on the magnetic core. It is measured in ampere-turns per meter. The vertical axis represents the **magnetic flux density B** , which is the amount of magnetic flux developed per unit cross-sectional area of the material, measured in tesla. As the magnetizing current in the coil is varied from positive to negative and back again, the value of H changes, and the corresponding value of B traces the hysteresis loop.

To understand how the loop is formed, begin from the origin when the material is initially unmagnetized. As the magnetizing force H is increased in the positive direction, the flux density B rises along the **dotted initial magnetization curve** until it reaches point **a**, where the material attains **positive saturation**. At this point, nearly all magnetic domains inside the material are aligned in the direction of the applied field, and further increase in H produces negligible increase in B . The flux density here is the maximum positive flux density, denoted as $+B_{max}$. When the magnetizing force is now reduced back to zero, the flux density does not return to zero. Instead, it follows the blue curve and reaches point **b**, where some magnetism remains in the material even though $H = 0$. This residual flux density is called the **retentivity** or **remanence**, denoted by $+B_r$. It represents the ability of the material to retain magnetization after the external magnetizing force is removed.

To reduce this residual magnetism to zero, a magnetizing force must be applied in the opposite direction. As negative H is applied, the flux density decreases and becomes zero at point **c**. The magnitude of reverse magnetizing force required to bring the residual flux density to zero is called the **coercive force** or **coercivity**, denoted by $-H_c$. This indicates the resistance of the material to demagnetization. If the negative magnetizing force is increased further, the material reaches point **f**, corresponding to **negative saturation**, where the flux density becomes $-B_{max}$. At this stage, all domains are aligned in the opposite direction.

Now when the magnetizing force is reduced from negative saturation back toward zero, the flux density again lags and does not return along the original path. Instead, it follows the purple curve and reaches point **e**, where a residual negative flux density remains. This is the negative remanent flux density $-B_r$. To bring this negative residual flux density back to zero, a positive magnetizing force must be applied. As H increases positively, the flux density becomes zero at point **d**, where the required positive coercive force is $+H_c$. Further increase in positive magnetizing force returns the material to point **a**, completing the loop.

Thus, the closed path **a** $\rightarrow$ **b** $\rightarrow$ **c** $\rightarrow$ **f** $\rightarrow$ **e** $\rightarrow$ **d** $\rightarrow$ **a** forms the hysteresis loop. The loop exists because the magnetic domains in the material require energy to rotate and realign whenever the magnetic field changes direction. **The area enclosed by the hysteresis loop represents the hysteresis energy loss per cycle per unit volume of the magnetic material.** A larger loop area indicates greater hysteresis loss, while a narrower loop indicates lower hysteresis loss and better magnetic efficiency. Physically, the “forces” involved in the graph are not mechanical forces but magnetic quantities. The **magnetizing force H** is the externally applied magnetic field generated by electric current through a coil, while the **flux density B** is the magnetic response of the material due to domain alignment. The loop is experimentally obtained by repeatedly magnetizing and demagnetizing a ferromagnetic specimen using alternating current and measuring the corresponding values of H and B .

Non-linear Magnetic Circuits

A magnetic circuit is said to be **non-linear** when the relationship between magnetic flux density (B) and magnetic field intensity (H) is not linear.

Cause of Non-linearity

This occurs due to **magnetic saturation** of the material.

Explanation

- Initially, $B \propto H$ (linear region)
 - As H increases, the material approaches saturation
 - Beyond a certain point, large increases in H produce very small increases in B
- This behavior is represented by the **B-H Curve**.

Implications

- Permeability (μ) is not constant
- Reluctance varies with flux
- Superposition does not apply
- Analysis becomes complex

Practical Consequences

- Magnetic devices (transformers, motors) must be designed to avoid deep saturation
- Non-linear analysis methods (graphical or numerical) are required

Eddy Current Loss

In a magnetic circuit, **eddy current loss** refers to the power dissipated in the magnetic core due to circulating currents induced within the core material itself when it is subjected to a time-varying magnetic flux. According to Faraday's law of electromagnetic induction, whenever the magnetic flux linking a conducting material changes with time, an electromotive force is induced in that material. Since magnetic cores are generally made of conducting ferromagnetic materials such as iron or steel, this induced emf drives circulating currents within the body of the core. These circulating currents are called eddy currents because they flow in closed loops resembling swirling eddies in water. As these currents pass through the electrical resistance of the core material, energy is converted into heat, leading to undesirable power loss and temperature rise in the magnetic circuit. Eddy current loss increases with the square of both the flux density and the supply frequency, and it also depends on the thickness of the core material. To minimize this loss, magnetic cores used in transformers, motors, and other AC magnetic devices are laminated into thin insulated sheets rather than made as a solid block. The lamination increases the resistance to the flow of eddy currents and significantly reduces their magnitude, thereby reducing the associated heating and energy loss.

Hysteresis Loss

Hysteresis loss is the energy dissipated in the magnetic material during each cycle of magnetization when the magnetic flux in the core repeatedly changes direction. Ferromagnetic materials are composed of numerous microscopic magnetic domains, each acting like a tiny magnet. When an external magnetizing force is applied, these domains align themselves in the direction of the magnetic field. However, due to internal molecular friction and domain resistance, the domains do not realign instantly or completely without expending energy. During every cycle of alternating magnetization, some energy is required to continuously reverse the orientation of these domains, and this energy is lost in the form of heat. The phenomenon responsible for this loss is known as magnetic hysteresis, which is represented graphically by the hysteresis loop of the material. The area enclosed by the hysteresis loop indicates the amount of energy lost per cycle per unit volume of the material. Hysteresis loss depends on the frequency of magnetization, the volume of the core, the maximum flux density, and the magnetic properties of the core material. Materials with narrow hysteresis loops, such as silicon steel, are preferred in magnetic circuits because they require less energy for repeated magnetization reversal and therefore exhibit lower hysteresis loss.

Transformer

INTRODUCTION

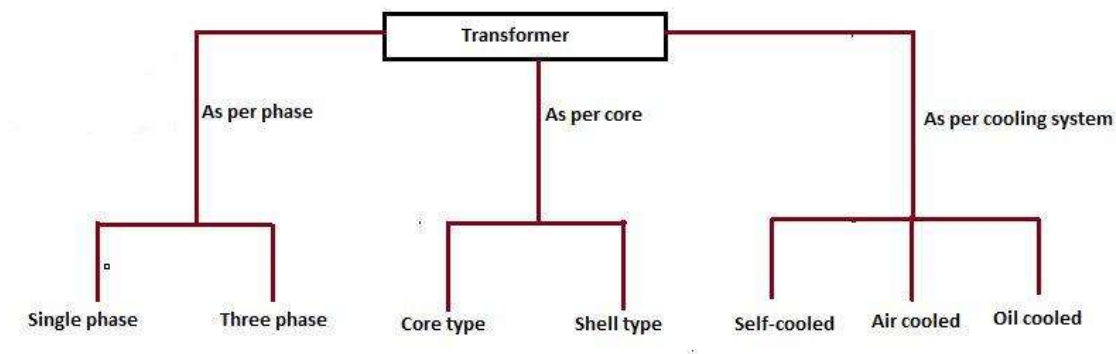
- A TRANSFORMER is a device that transfers electrical energy from one circuit to another by electromagnetic induction (transformer action).
- The electrical energy is always transferred without a change in frequency, but may involve changes in magnitudes of voltage and current.
- Transformer works on the principle of electromagnetic induction, it must be used with an input source voltage that varies in amplitude.

Types of Transformer:

Transformers are used in various fields like power generation grid, distribution sector, transmission and electric energy consumption. There are various types of transformers which are classified based on the following factors;

- Working voltage range.
- The medium used in the core.
- Winding arrangement.
- Installation location.

Types of Transformer



Based on Voltage Levels

Commonly used transformer type, depending upon voltage they are classified as:

- **Step-up Transformer:** They are used between the power generator and the power grid. The secondary output voltage is higher than the input voltage.
- **Step down Transformer:** These transformers are used to convert high voltage primary supply to low voltage secondary output.

Based on the Medium of Core Used

In a transformer, we will find different types of cores that are used.

- **Air core Transformer:** The flux linkage between primary and secondary winding is through the air. The coil or windings wound on the non-magnetic strip.
- **Iron core Transformer:** Windings are wound on multiple iron plates stacked together, which provides a perfect linkage path to generate flux.

Based on the Winding Arrangement

- **Autotransformer:** It will have only one winding wound over a laminated core. The primary and secondary share the same coil.

Based on Install Location

- **Power Transformer:** It is used at power generation stations as they are suitable for high voltage application
- **Distribution Transformer:** Mostly used at distribution lanes in domestic purposes. They are designed for carrying low voltages. It is very easy to install and characterized by low magnetic losses.
- **Measurement Transformers:** These are further classified. They are mainly used for measuring voltage, current, power.
- **Protection Transformers:** They are used for component protection purposes. In circuits some components must be protected from voltage fluctuation etc. protection transformers ensure component protection.

The major parts of a single-phase transformer:

1. Core

The core acts as a support to the winding in the transformer. It also provides a low reluctance path to the flow of magnetic flux. The winding is wound on the core as shown in the picture. It is made up of a laminated soft iron core in order to reduce the losses in a transformer. The factors such as operating voltage, current, power etc decide core composition. The core diameter is directly proportional to copper losses and inversely proportional to iron losses.

2. Windings

Windings are the set of copper wires wound over the transformer core. Copper wires are used due to:

- High conductivity of copper. This minimizes the loss in a transformer. since conductivity increases, resistance to current flow decreases.
- High ductility of copper. Ductility is the property of metals that they can be made into very thin wires.

There are mainly two types of windings. Primary windings and secondary windings.

- Primary winding: The set of turns of windings to which supply current is feed.
- Secondary winding: the set of turns of winding from which output is taken.

The primary and secondary windings are insulated from each other using insulation coating agents.



3. Insulation Agents

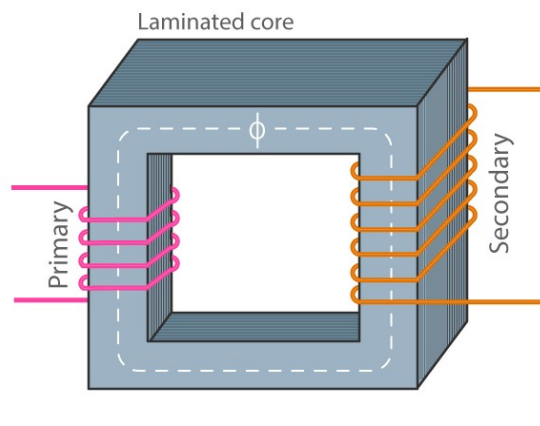
Insulation is necessary for transformers to separate windings from shorting the circuit and thus facilitating the mutual induction. Insulation agents have influence in durability and the stability of a transformer.

Following are used as an insulation medium in a transformer:

- Insulating oil
- Insulating tape
- Insulating paper
- Wood-based lamination

Working Principle of a Transformer:

BASIC WORKING PRINCIPLE OF TRANSFORMER A transformer can be defined as a static device which helps in the transformation of electric power in one circuit to electric power of the same frequency in another circuit. The voltage can be raised or lowered in a circuit, but with a proportional increase or decrease in the current ratings. The main principle of operation of a transformer is mutual inductance between two circuits which is linked by a common magnetic flux. A basic transformer consists of two coils that are electrically separate and inductive, but are magnetically linked through a path of reluctance. The working principle of the transformer can be understood from the figure below.



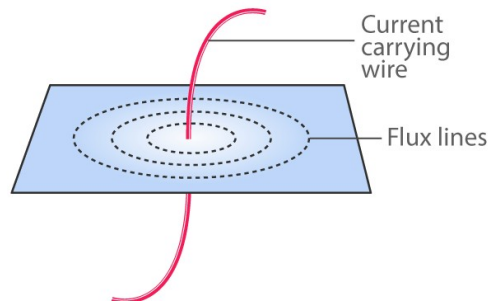
As shown above the transformer has primary and secondary windings. The core laminations are joined in the form of strips in between the strips you can see that there are some narrow gaps right through the cross-section of the core. These staggered joints are said to be 'imbricated'. Both the coils have high mutual inductance. A mutual electro-motive force is induced in the transformer from the alternating flux that is set up in the laminated core, due to the coil that is connected to a source of alternating voltage. Most of the alternating flux developed by this coil is linked with the other coil and thus produces the mutual induced electromotive force. The so produced electro-motive force can be explained with the help of Faraday's laws of Electromagnetic Induction as $e = M \cdot dI/dt$. If the second coil circuit is closed, a current flows in it and thus electrical energy is transferred magnetically from the first to the second coil.

The alternating current supply is given to the first coil and hence it can be called as the primary winding. The energy is drawn out from the second coil and thus can be called as the secondary winding.

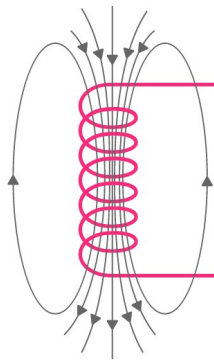


In short, a transformer carries the operations shown below:

- Transfer of electric power from one circuit to another.
- Transfer of electric power without any change in frequency.
- Transfer with the principle of electromagnetic induction.
- The two electrical circuits are linked by mutual induction.



The figure shows the formation of magnetic flux lines around a current-carrying wire. The normal of the plane containing the flux lines are parallel to normal of a cross-section of a wire.



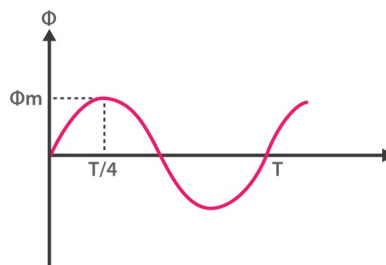
The figure shows the formation of varying magnetic flux lines around a wire-wound. The interesting part is that reverse is also true, when a magnetic flux line fluctuates around a piece of wire, a current will be induced in it.

Ideal Transformer

The ideal transformer has no losses that means there is no magnetic leakage and ohmic resistance in its windings. There won't be any core losses or I^2R .

The current passing through the primary coil creates a magnetic field. The primary and secondary coils are wrapped around a core of very high magnetic permeability, such as iron, so that most of the magnetic flux passes through both the primary and secondary coils.

EMF Equation of Transformer:



N_1 – number of turns in primary.

N_2 – number of turns in secondary.

Φ_m – maximum flux in weber (Wb).

T – time period. Time is taken for 1 cycle.

The flux formed is a sinusoidal wave. It rises to a maximum value Φ_m and decreases to negative maximum Φ_m . So, flux reaches a maximum in one-quarter of a cycle. The time taken is equal to $T/4$.

Average rate of change of flux = $\Phi_m/(T/4) = 4*f\Phi_m$

Where f = frequency

$T = 1/f$

Induced emf per turn = rate of change of flux per turn

Form factor = RMS value / average value

RMS value = $1.11 * (4*f\Phi_m) = 4.44 f\Phi_m$ [form factor of sine wave is 1.11]

RMS value of emf induced in winding = RMS value of emf per turn * no of turns

Primary Winding

RMS value of induced EMF = $E_1 = 4.44 f\Phi_m * N_1$

Secondary winding:

RMS value of induced EMF = $E_2 = 4.44 f\Phi_m * N_2$

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = 4.44f\Phi_m$$

This is the emf equation of the transformer.

For an ideal transformer at no load condition,

E_1 = supply voltage on the primary winding.

E_2 = terminal voltage (theoretical or calculated) on the secondary winding.

Voltage Transformation Ratio

$$\frac{E_1}{N_1} = \frac{E_2}{N_2} = k$$

K is called the voltage transformation ratio, which is a constant.

Case1: if $N_2 > N_1$, $K > 1$ it is called a step-up transformer.

Case 2: if $N_2 < N_1$, $K < 1$ it is called a step-down transformer.

Applications of Transformer

- The transformer transmits electrical energy through wires over long distances.
- Transformers with multiple secondary's are used in radio and TV receivers which require several different voltages.



- Transformers are used as voltage regulators.

Voltage Transformation Ratio.

The ratio of secondary voltage to primary voltage is known as the voltage transformation ratio and is designated by letter K. i.e. Voltage transformation ratio,

$$K = V_2/V_1 = E_2/E_1 = N_2/N_1$$

Current Ratio.

The ratio of secondary current to primary current is known as current ratio and is reciprocal of voltage transformation ratio.

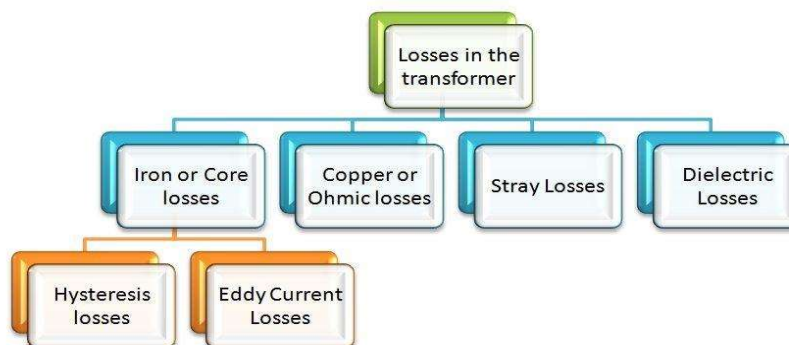
$$\frac{I_2}{I_1}$$



Types of Losses in a Transformer

There are various types of losses in the transformer such as iron loss, copper loss, hysteresis loss, eddy current loss, stray loss, and dielectric loss. The hysteresis losses occur because of the variation of the magnetization in the core of the transformer and the copper loss occurs because of the transformer winding resistance.

The various types of losses are explained below in detail.



Types of Losses in the Transformer

Iron Losses

Iron losses are caused by the alternating flux in the core of the transformer as this loss occurs in the core it is also known as **Core loss**. Iron loss is further divided into hysteresis and eddy current loss.

Hysteresis Loss

The core of the transformer is subjected to an alternating magnetizing force, and for each cycle of emf, a hysteresis loop is traced out. Power is dissipated in the form of heat known as hysteresis loss and given by the equation shown below:

$$P_h = K\eta B_{\max}^{1.6} f V \text{ watts}$$

Where

- $K\eta$ is a proportionality constant which depends upon the volume and quality of the material of the core used in the transformer,
- f is the supply frequency,
- $B_{\max}$ is the maximum or peak value of the flux density.

The iron or core losses can be minimized by using silicon steel material for the construction of the core of the transformer.

Eddy Current Loss

When the flux links with a closed circuit, an emf is induced in the circuit and the current flows, the value of the current depends upon the amount of emf around the circuit and the resistance of the circuit.

Since the core is made of conducting material, these EMFs circulate currents within the body of the material. These circulating currents are called Eddy Currents. They will occur when the conductor experiences a changing magnetic field. As these currents are not responsible for doing any useful work, and it produces a loss (I^2R loss) in the magnetic material known as an Eddy Current Loss. The eddy current loss is minimized by making the core with thin laminations. The equation of the eddy current loss is given as:

$$P_e = K_e B_m^2 t^2 f^2 V \text{ watts}$$

Where,

- K_e – coefficient of eddy current. Its value depends upon the nature of magnetic material like volume and resistivity of core material, the thickness of laminations
- B_m – maximum value of flux density in wb/m^2

- T – thickness of lamination in meters
- F – frequency of reversal of the magnetic field in Hz
- V – the volume of magnetic material in m³

Copper Loss Or Ohmic Loss

These losses occur due to ohmic resistance of the transformer windings. If I_1 and I_2 are the primary and the secondary current. R_1 and R_2 are the resistance of primary and secondary winding then the copper losses occurring in the primary and secondary winding will be $I_1^2 R_1$ and $I_2^2 R_2$ respectively.

Therefore, the total copper losses will be

$$P_c = I_1^2 R_1 + I_2^2 R_2$$

These losses varied according to the load and known hence it is also known as variable losses. Copper losses vary as the square of the load current.

Stray Loss

The occurrence of these stray losses is due to the presence of leakage field. The percentage of these losses are very small as compared to the iron and copper losses so they can be neglected.

Dielectric Loss

Dielectric loss occurs in the insulating material of the transformer that is in the oil of the transformer, or in the solid insulations. When the oil gets deteriorated or the solid insulation gets damaged, or its quality decreases, and because of this, the efficiency of the transformer gets affected.



Transformer Efficiency

Transformer efficiency (η) can be explained as the ratio of the output power to the input power.

$$\eta \triangleq \frac{\text{output power}}{\text{input power}}$$
$$= \frac{\text{output power}}{\text{output power} + \text{copper loss} + \text{iron loss}} \text{ pu}$$

Therefore the per unit efficiency at load current I_2 and power factor $\cos \Phi_2$ will be

$$\eta = \frac{V_2 I_2 \cos \Phi_2}{V_2 I_2 \cos \Phi_2 + I_2^2 \cdot R_{e2} + P_i} \text{ pu}$$

And the per unit efficiency at full load is

$$\eta_{fl} = \frac{V_2 I_{2fl} \cos \Phi_2}{V_2 I_{2fl} \cos \Phi_2 + I_{2fl}^2 \cdot R_{e2} + P_i}$$

If $S_{2fl} = (VA)_{2fl} = V_2 I_{2fl}$ = full- load VA = rated VA

Then

$$\eta_{fl} = \frac{S_2 \cos \Phi_2}{S_2 \cos \Phi_2 + P_{eff} R_{e2} + P_i}$$

As we know that in a transformer there is no rotational part so there are no rotational losses such as windings and frictional losses in a rotating machine. Therefore, we can obtain a maximum efficiency as high as 99% in a well-designed transformer.

Transformer Voltage Regulation

Voltage Regulation of single-phase transformers is the percentage (or per unit value) change in its secondary terminal voltage compared to its original no-load voltage under varying secondary load conditions. In other words, regulation determines the variation in secondary terminal voltage which occurs inside the transformer as a result of variations in the transformers connected load thereby affecting its performance and efficiency if these losses are high and the secondary voltage becomes too low.

A transformer's voltage regulation change between its secondary terminal voltage from a no-load condition when $I_L = 0$, (open circuit) to a fully-loaded condition when $I_L = I_{MAX}$ (maximum current) for a constant primary voltage is given as:

Transformer Voltage Regulation

$$\text{Regulation} = \frac{\text{Change in Output Voltage}}{\text{No-load Output Voltage}}$$

$$\therefore \text{Regulation} = \frac{V_{(\text{no-load})} - V_{(\text{full-load})}}{V_{(\text{no-load})}}$$

Transformer Voltage Regulation as a Percentage Change

$$\% \text{ Regulation} = \frac{V_{\text{no-load}} - V_{\text{full-load}}}{V_{\text{full-load}}} \times 100$$

Where

$V_{\text{no-load}}$ is the no-load voltage and
 $V_{\text{full-load}}$ is the full-load voltage.

